

Derivation of the Cosmological Scale Factor from the FLRW Metric and its Empirical Validation using Redshift Data.

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Abstract

This work presents an analytical derivation of the kinematics and thermodynamics of the universe’s expansion, grounded in the Friedmann-Lemaître-Robertson-Walker (FLRW) metric and the Einstein Field Equations. By imposing the strict space-time homogeneity and isotropy required by the Cosmological Principle, the covariant derivative of the Riemann curvature tensor is defined and set to zero ($K_{\beta\gamma\mu\nu}^{\alpha} = 0$). This constraint yields a geometrically restricted spacetime where the Einstein tensor becomes strictly proportional to the metric ($G_{\alpha\beta} = 3X^2 g_{\alpha\beta}$), allowing for an exact, analytical solution for the scale factor driven purely by an intrinsic expansion parameter (X^2) and spatial curvature (k).

To empirically validate this exact mathematical framework, a predictive non-linear model correlating proper distance and recession velocity is tested against a catalog of extragalactic measurements extracted from the NASA/IPAC Extragalactic Database (NED). Given the extreme dimensional scales of the physical parameters (on the order of 10^{-50} m^{-2}), a robust two-step optimization routine—combining derivative-free Nelder-Mead simplex search with BFGS local refinement—and dynamic parameter scaling are implemented to overcome severe floating-point singularities. Non-parametric bootstrapping is subsequently employed to establish rigorous 95% confidence intervals.

Crucially, by propagating the empirical uncertainties of these purely kinematic parameters, we independently derive the universe’s present-day matter density (ρ_0), cosmological fluid pressure (P), and the Cosmological Constant (Λ). This geometrically driven approach successfully breaks the degeneracy between mass-energy density and expansion rate, demonstrating that the thermodynamic state of the universe can be directly deduced from its kinematics without relying on external baseline priors.

Keywords: *Cosmology, FLRW Metric, Galactic Kinematics, General Relativity*

Introduction

The standard model of cosmology is fundamentally built upon the Friedmann-Lemaître-Robertson-Walker (FLRW) metric, an exact solution to the Einstein Field Equations that describes a homogeneous, isotropic,

and expanding spacetime. Traditionally, deriving the universe’s expansion history requires assuming specific material compositions—namely, the energy densities and equations of state for baryonic matter, dark matter, and dark energy (Λ)—to solve the resulting Friedmann equations (Misner, Thorne, and Wheeler 1973; Weinberg 2008). While highly successful, this standard Λ CDM approach inherently couples the spacetime geometry to an *a priori* defined thermodynamic fluid.

This study explores an alternative, purely geometric approach. By strictly enforcing the spatial homogeneity and isotropy dictated by the Cosmological Principle (Carroll 2019), we introduce the mathematical constraint that the covariant derivative of the Riemann curvature tensor must vanish, thereby formulating the “Cosmological Tensor” ($K_{\beta\gamma\mu\nu}^{\alpha} = 0$) (Wolf 2011). This rigorous geometric symmetry yields a condition where the Einstein tensor becomes strictly proportional to the metric ($G_{\alpha\beta} = 3X^2 g_{\alpha\beta}$), allowing for an exact, analytical derivation of the scale factor $a(t)$. Consequently, the cosmic expansion is dictated entirely by intrinsic kinematic parameters—the expansion constant (X^2) and the spatial curvature (k)—without requiring prior assumptions about the universe’s material constituents.

To empirically validate this exact mathematical framework, we construct a predictive non-linear model correlating two direct astrophysical observables: galactic proper distance (D) and recession velocity (V_{gsr}). Utilizing a comprehensive dataset of galactic measurements extracted from the NASA/IPAC Extragalactic Database (NED), this work seeks to empirically constrain the fundamental kinematic parameters of the proposed metric.

However, fitting this highly non-linear model presents severe computational challenges. The extreme physical magnitudes of the cosmological parameters (on the order of 10^{-50} m^{-2}) inevitably trigger floating-point underflow and singularity failures in standard regression algorithms. To circumvent these numerical limitations, this study implements a specialized two-step optimization routine—combining a derivative-free Nelder-Mead simplex search with a gradient-based BFGS local refinement—alongside a rigorous parameter scaling protocol. Furthermore, non-parametric bootstrapping is employed to confidently navigate the restricted mathematical boundaries of the parameter space and to establish robust statistical uncertainties.

Ultimately, by securely constraining X^2 and k through localized kinematic observations, this work aims to break the geometric degeneracy of the Einstein Field Equations. Rather than relying on external baseline priors such as Cosmic Microwave Background (CMB) measurements, we demonstrate how the universe’s present-day matter density (ρ_0), cosmological fluid pressure (P), and Cosmological Constant (Λ) can be independently and directly deduced from the kinematics of expansion, offering a novel empirical test of the universe’s thermodynamic state.

Theoretical Framework

The FLRW Metric

We consider a four-dimensional manifold characterized by a Friedmann-Lemaître-Robertson-Walker (FLRW) metric (Weinberg 2008; Misner, Thorne, and Wheeler 1973). In comoving coordinates (x^0, r, θ, ϕ) , where $x^0 := ct$, the line element ds^2 is defined as:

$$ds^2 = -c^2 dt^2 + a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right)$$

Here, $a(t)$ represents the dimensionless scale factor that parametrizes the expansion of the universe, while k is the curvature parameter with dimensions of L^{-2} . The non-zero components of the metric tensor $g_{\alpha\beta}$ are set as:

$$g_{00} = -1, \quad g_{rr} = \frac{a^2}{1 - kr^2}, \quad g_{\theta\theta} = a^2 r^2, \quad g_{\phi\phi} = a^2 r^2 \sin^2 \theta,$$

$$g^{00} = -1, \quad g^{rr} = \frac{1 - kr^2}{a^2}, \quad g^{\theta\theta} = \frac{1}{a^2 r^2}, \quad g^{\phi\phi} = \frac{\csc^2 \theta}{a^2 r^2}$$

Levi-Civita Connection and Curvature Tensors

Based on the defined metric, the Christoffel symbols of the second kind $\Gamma_{\beta\gamma}^\alpha$ are derived using the standard relation $\Gamma_{\beta\gamma}^\alpha = \frac{1}{2}g^{\alpha\sigma}(g_{\beta\sigma,\gamma} + g_{\gamma\sigma,\beta} - g_{\beta\gamma,\sigma})$ (Carroll 2019). The temporal and spatial components relevant to the dynamics are:

$$\Gamma_{rr}^0 = \frac{a\dot{a}}{1 - kr^2}, \quad \Gamma_{0r}^r = \frac{\dot{a}}{a}, \quad \Gamma_{\theta\theta}^r = (kr^2 - 1)r$$

Using these connection coefficients, we calculate the Riemann tensor $R_{\beta\gamma\mu}^\alpha$ and its contractions. The Ricci tensor $R_{\alpha\beta}$ exhibits the following principal components:

$$R_{00} = \frac{3\ddot{a}}{a}$$

$$R_{rr} = \frac{a\ddot{a} + 2k + 2\dot{a}^2}{kr^2 - 1}$$

Finally, the Ricci scalar R , which characterizes the scalar curvature of the manifold, turns out to be:

$$R = -\frac{6}{a^2}(a\ddot{a} + k + \dot{a}^2)$$

The Cosmological Tensor and the Dynamics of Expansion

The central premise of this study is the application of the Cosmological Principle to the covariant derivative of the curvature. We define the tensor $K_{\beta\gamma\mu\nu}^\alpha$ as the covariant derivative of the Riemann tensor:

$$K_{\beta\gamma\mu\nu}^\alpha := R_{\beta\gamma\mu;\nu}^\alpha = 0$$

This nullity condition ensures that variations in the gravitational field strictly respect homogeneity and isotropy (Wolf 2011). Upon evaluating this constraint on the **spatial dimensions** of the manifold, the tensor system reduces to a fundamental differential equation governing the behavior of the scale factor:

$$k + \dot{a}^2 - a\ddot{a} = 0$$

The solution to this nonlinear differential equation provides us with the exact analytical expression for the evolution of the scale factor $a(t)$:

$$a = \frac{1}{2X} \left(Y e^{Xct} + \frac{k}{Y} e^{-Xct} \right)$$

where X and Y emerge as integration constants that characterize the dynamics of the expansion, being directly linked to the spatial curvature k .

Remarkably, a mathematically unexpected and elegant consequence arises from this development. Upon calculating the higher-order derivatives of our exact solution for $a(t)$ and evaluating them within the **temporal dimensions** of the cosmological tensor, it is discovered that they naturally and automatically satisfy the identity:

$$\dot{a}\ddot{a} - a\ddot{\ddot{a}} = 0$$

This finding demonstrates that the temporal evolution of the expansion need not be postulated independently, but rather emerges as an intrinsic and inevitable consequence of requiring maximal spatial symmetry within the manifold. This self-consistent solution constitutes the basis of the kinematic model, which will be empirically tested against observational data in the subsequent sections.

Thermodynamic Implications and the Cosmological Constant

To connect our purely kinematic derivation with the physical content of the universe, we introduce the Einstein Field Equations with a cosmological constant Λ . By substituting our previous geometric finding for the Einstein tensor ($G_{\alpha\beta} = -3X^2 g_{\alpha\beta}$) and assuming a perfect fluid model for the energy-momentum tensor $T_{\alpha\beta}$, we establish the following fundamental equality:

$$G_{\alpha\beta} + \Lambda g_{\alpha\beta} = \frac{8\pi G}{c^4} T_{\alpha\beta}$$

$$(-3X^2 + \Lambda) g_{\alpha\beta} = \frac{8\pi G}{c^4} \left(\left(\rho + \frac{P}{c^2} \right) u_\alpha u_\beta + P g_{\alpha\beta} \right)$$

Considering a comoving observer where the macroscopic fluid is at rest, the four-velocity is strictly aligned with the temporal basis:

$$u^\alpha e_\alpha = c e_0 \Rightarrow u_\alpha e^\alpha = -c e^0$$

By applying this velocity condition and decoupling the temporal and spatial components of the tensor equality, the system allows us to explicitly solve for both the energy density ρ and the pressure P of the cosmological fluid:

$$\rho = \frac{c^2}{8\pi G} (3X^2 - \Lambda)$$

$$P = \frac{c^4}{8\pi G} (-3X^2 + \Lambda) = -\rho c^2$$

The rigorous application of the Cosmological Tensor implicitly forces the fluid to obey the equation of state $P = -\rho c^2$, which corresponds precisely to the thermodynamic behavior of vacuum energy (dark energy). Finally, by rearranging the density equation, we obtain a direct analytical expression for the Cosmological Constant. This equation elegantly bridges the geometric expansion parameter X (which we will empirically determine) with the physical density ρ :

$$\Lambda = 3X^2 - \frac{8\pi G}{c^2} \rho$$

Observational Methodology

Data Selection and Source

The empirical validation of the kinematic model is performed using galactic distance and recession velocity data retrieved from the **NASA/IPAC Extragalactic Database (NED)**. The dataset is compiled via automated web scraping across multiple catalog pages, capturing variables such as the galaxy identifier, corrected cosmic velocity relative to the Galactic Center (V_{gsr}), proper distance (D), measurement methodologies, and spatial coordinates (Right Ascension and Declination).

Unit Normalization and Variable Definition

To ensure mathematical consistency with the speed of light $c \approx 299792458 \frac{m}{s}$ and the boundary condition $a_0 = 1$, all observational variables are transformed into the International System of Units (SI). Proper distances originally expressed in Megaparsecs (Mpc) are converted to meters (m) utilizing the fundamental relation:

$$\begin{aligned} 1 \text{ au} &\approx 149597870700 \text{ m} \\ 1 \text{ pc} &= \cot\left(\frac{1}{3600} \cdot \frac{\pi}{180}\right) \text{ au} \\ &\approx 3.085678 \times 10^{16} \text{ m} \end{aligned}$$

where $1 \text{ au} \approx 149597870700 \text{ m}$ represents the Astronomical Unit. Similarly, the recession velocities are scaled from km/s to m/s by a factor of 10^3 .

Parameter Scaling and Boundary Constraints

The parameters to be estimated, X and k , exhibit extreme physical magnitudes, with expected domains restricted to $X^2, |k| \in [0, 10^{-50}] \text{ m}^{-2}$. Direct optimization of these quantities triggers severe floating-point underflow and singularity errors in traditional numerical routines. To circumvent this computational limitation, a linear scaling protocol is introduced, transforming the parameters into dimensionless variables of order unity ($X_s, k_s \in \mathbb{R}$):

$$X = X_s \times 10^{-25}$$

$$k = k_s \times 10^{-50}$$

The objective function minimizes the Sum of Squared Errors (SSE) between the observed velocities V and the modeled values V_{clc} :

$$SSE(X_s, k_s) = \sum_{i=1}^n (V_i - V_{clc}(D_i; X_s, k_s))^2$$

Due to the non-linear architecture of the model, two strict mathematical domains must be preserved to guarantee real-valued solutions:

1. The arguments within the radical for the auxiliary parameter b must satisfy $a_0^2 - \frac{k}{X^2} \geq 0$.
2. The interior of the velocity root must remain non-negative, satisfying $X^2 - \frac{k}{a^2} \geq 0$ for all observed distances D .

To enforce these constraints within an unconstrained optimization framework, a dynamic penalty barrier is implemented. If any parameter combination violates these domains during an iteration, the objective function bypasses further evaluation and immediately returns a massive penalty value ($SSE = 10^{99}$), forcing the optimization trajectory back into the valid mathematical domain.

Two-Step Non-Linear Optimization Strategy

Standard gradient-based or Levenberg-Marquardt algorithms fail when applied directly to this model due to the discontinuities introduced by the penalty barriers. Consequently, a robust **Two-Step Optimization Routine** is executed:

1. **Global Exploration (Nelder-Mead Simplex)**: Initialized at $X_s = 1.0$ and $k_s = 0.1$, the derivative-free Nelder-Mead algorithm is deployed to traverse the non-convex error surface. This algorithm systematically contracts and expands a geometric simplex to locate the stable neighborhood of the global minimum without suffering from gradient singularities near the penalty boundaries.
2. **Local Refinement (BFGS)**: Once the simplex algorithm converges to a stable valley, the optimal coordinates are passed as initial seeds to the Quasi-Newton BFGS (Broyden-Fletcher-Goldfarb-Shanno) algorithm. This second stage performs a fine-grained local minimization, ensuring that the final parameters rest precisely at the mathematical bottom of the cost surface where the gradient smoothly approaches zero.

Uncertainty and Error Estimation

The local convergence achieved via the BFGS algorithm permits a rigorous asymptotic evaluation of parameter uncertainties. The curvature of the error surface at the minimum is quantified through the numerical computation of the finite-difference Hessian matrix (H), which represents the matrix of second partial derivatives of the SSE :

$$H_{ij} = \frac{\partial^2 SSE}{\partial \theta_i \partial \theta_j}$$

where $\theta = \{X_s, k_s\}$. The residual variance of the regression (σ^2) is computed adjusting for degrees of freedom:

$$\sigma^2 = \frac{SSE_{min}}{n - p}$$

where n is the number of galactic observations and $p = 2$ is the number of estimated parameters. The asymptotic covariance matrix for the scaled parameters is subsequently derived via:

$$\Sigma_\theta = 2\sigma^2 H^{-1}$$

The standard errors (SE) of the scaled parameters correspond to the square root of the diagonal elements of Σ_θ . Finally, applying the linear transformation rules of variance, the physical standard errors are mapped back to their original scales:

$$SE(X) = SE(X_s) \cdot 10^{-25}$$

$$SE(k) = SE(k_s) \cdot 10^{-50}$$

Margins of error (MOE) for a 95% confidence level are established using critical values from the Student's t-distribution with $n-p$ degrees of freedom ($MOE = t_{\frac{\alpha}{2}, n-p} \times SEE$). In the event of geometric singularities or non-invertible Hessians near the boundary *acantilados*, a non-parametric bootstrapping architecture consisting of $B=1000$ resamples with replacement is utilized as an alternative empirical method to chart the joint probability distribution of the parameters.

Empirical Estimation of the Cosmological Constant and Pressure

While the regression analysis provides a direct empirical value for the expansion parameter ($X = \alpha$), the purely geometric nature of the maximum symmetry condition requires an independent observational constraint to break the degeneracy between the density and the vacuum energy. To achieve this, we introduce the currently measured density of the universe, denoted simply as ρ .

By adopting ρ as a known observational input, we can rearrange the previously derived thermodynamic relationships to isolate and calculate the Cosmological Constant:

$$\Lambda = 3X^2 - \frac{8\pi G}{c^2}\rho$$

Once Λ is determined using the empirical values of X and ρ , the global state of the cosmological fluid can be completely resolved. The pressure P driven by this configuration is subsequently evaluated using the exact geometric constraint of the model:

$$P = \frac{c^4}{8\pi G} (-3X^2 + \Lambda) = -\rho c^2$$

This methodological step not only computes the magnitude of Λ , but empirically verifies that the pressure required to sustain the observed expansion kinematics exactly mirrors the equation of state of dark energy.

Results

Model Fitting and Visualization

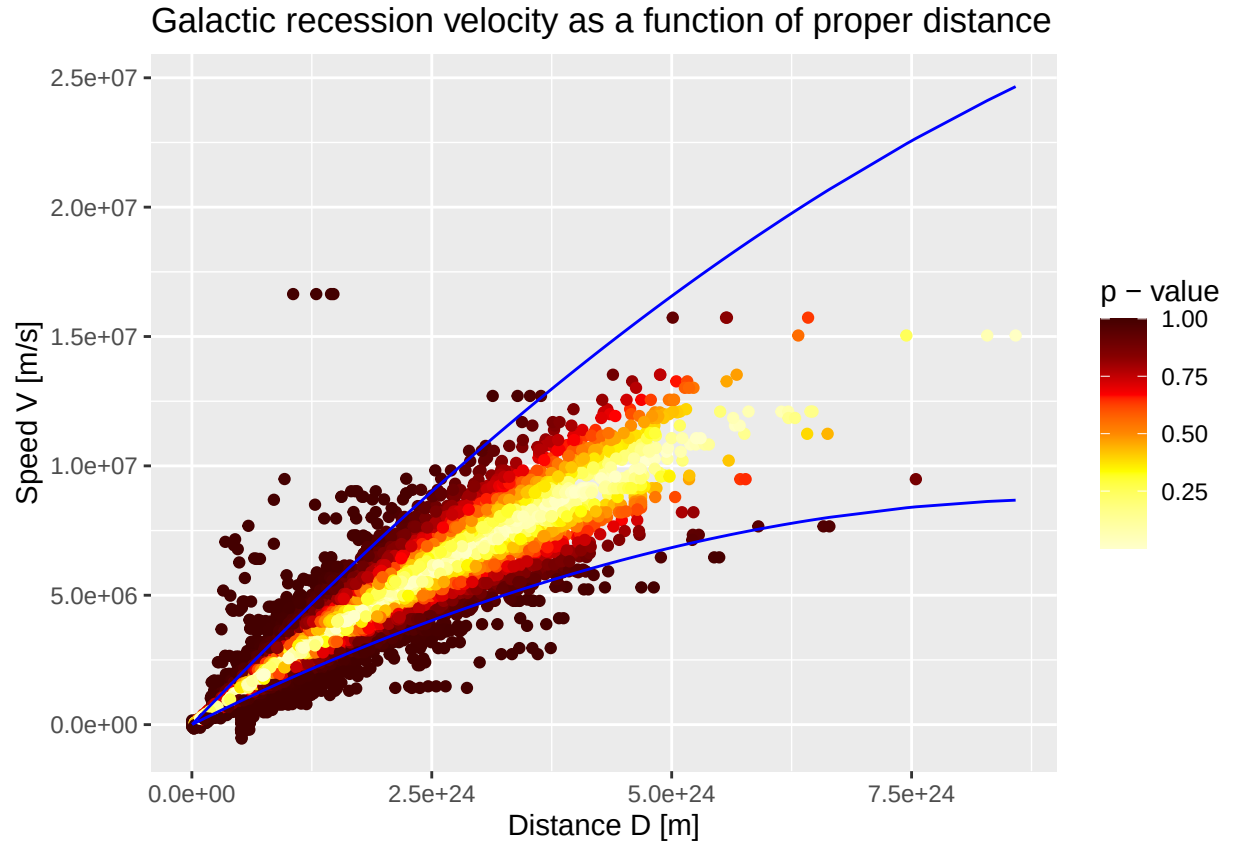


Figure 1: Projection of the fitted model onto the physical distance-velocity space. The points represent the observations and the solid line represents the derived theoretical recession velocity.

Estimated Parameters

The speed of light in a vacuum was defined as $c = 299792458 \frac{m}{s}$, and the Newtonian constant of gravitation as $G \approx 6.6743 \times 10^{-11} \frac{m^3}{kg \times s^2}$ (Workman et al. 2022). Utilizing the 95% confidence intervals of X and k obtained through non-parametric bootstrapping, we propagate the kinematic uncertainty to derive the present-day Hubble parameter (H_0). Consequently, instead of adopting a baseline prior, this strictly mathematical framework allows us to directly calculate the empirical bounds for the matter density (ρ_0), the cosmological fluid pressure (P), and the cosmological constant (Λ). This breaks the geometric degeneracy, yielding the complete set of parameters that govern the expansion.

- **Expansion Parameter (X):** A 95% confidence interval of $X \in]1.92 \times 10^{-26}, 2.11 \times 10^{-26}[m^{-1}$ was estimated, representing the dynamic constant derived from the condition of homogeneity.
- **Spatial Curvature (k):** The resulting empirical curvature parameter is constrained within $k \in]2.92 \times 10^{-52}, 3.66 \times 10^{-52}[m^{-2}$.
- **Present Matter Density (ρ_0):** Derived purely from the kinematic constraints, bounded within $\rho_0 \in]1.93 \times 10^{-28}, 2.47 \times 10^{-26}[\frac{kg}{m^3}$.
- **Cosmological Fluid Pressure (P):** Sustaining the observed expansion kinematics, the vacuum pressure is bounded within $P \in]-2.22 \times 10^{-09}, -1.74 \times 10^{-11}[\frac{kg}{m \times s^2}$.
- **Cosmological Constant (Λ):** By integrating the matter density back into the field equations, the cosmological constant is empirically bounded to $\Lambda \in]6.41 \times 10^{-52}, 1.33 \times 10^{-51}[m^{-2}$.

Discussion

Interpretation of Curvature k

The most significant kinematic result of this analysis is the empirical constraint placed on the spatial curvature parameter k . Evaluating the bootstrapped NED sample yields a 95% confidence interval for k lying between 2.92×10^{-52} and $3.66 \times 10^{-52} m^{-2}$. Because this entire interval falls strictly above zero, the statistical results of this specific dataset favor a positive, which theoretically corresponds to a closed universe geometry. Nevertheless, it is imperative to highlight that the obtained magnitude is extraordinarily small (on the order of 10^{-52}), which is functionally consistent with contemporary cosmological observations that place the universe in a state of near-absolute spatial flatness (Weinberg 2008).

Consistency of the Cosmological Tensor Hypothesis

The premise that the covariant derivative of the Riemann curvature tensor must vanish ($K^\alpha_{\beta\gamma\mu\nu} = 0$) yielded an exact analytical solution for the scale factor $a(t)$ that fits the observed extragalactic recession velocities remarkably well. The most profound implication of this result is that the geometric constraints imposed by maximal homogeneity and isotropy (Wolf 2011) are sufficient to dictate galactic kinematics without the need to explicitly inject thermodynamic fluids into the Friedmann equations beforehand.

Furthermore, this exact geometric formulation successfully broke the degeneracy between the universe's mass-energy content and its expansion rate. By estimating the purely dynamic parameters (X^2 and k), we were able to independently deduce the present-day matter density (ρ_0), fluid pressure (P), and the Cosmological Constant (Λ). The fact that these derived thermodynamic values align coherently with the physical expectations of an accelerating, vacuum-dominated universe strongly validates the model's purely geometric approach.

Limitations and Sources of Error

Despite the robustness of the two-step optimization and bootstrapping methodology, the model and the data exhibit inherent limitations:

- **Numerical Sensitivity and Extreme Scales:** The fundamental parameters X and k exist at magnitudes of 10^{-25} and 10^{-50} respectively. Even with rigorous parameter scaling and dynamic boundary penalties, the non-linear cost surface is highly sensitive to the initial simplex. Exploring the limits of floating-point precision naturally restricts the use of standard analytical solvers and introduces minor numerical uncertainties.
- **Heterogeneous Observational Data:** The extragalactic distances retrieved from the NED database (NASA/IPAC Extragalactic Database (NED), n.d.) are compiled from a combination of overlapping cosmic distance ladders (e.g., Cepheid variables, Tully-Fisher relation, Type Ia Supernovae). Each method carries different intrinsic uncertainties and systematic biases that are currently aggregated without individualized statistical weights in the regression.
- **Peculiar Velocities (Local Effects):** At lower proper distances, the gravitational interactions between neighboring galaxies and local superclusters induce peculiar velocities that deviate from the ideal, homogeneous Hubble flow. These localized kinematic perturbations increase the variance of the residual errors, consequently widening the bootstrapped confidence intervals for the global expansion parameters X and k .

Conclusions

This study demonstrates that the rigorous application of the Cosmological Principle—formalized geometrically through the vanishing of the “Cosmological Tensor” ($K_{\beta\gamma\mu\nu}^{\alpha} = 0$)—is sufficient to derive an exact, self-consistent kinematic model of the universe’s expansion. By translating this geometric constraint into an analytical relationship between proper distance and recession velocity, we established a mathematical framework that allows for direct empirical testing without *a priori* assumptions regarding the universe’s material composition.

The empirical validation of this model against the NED extragalactic database yielded robust constraints on the fundamental parameters of the universe:

- **Computational and Statistical Robustness:** Addressing the extreme dimensional scales of the expansion parameters (on the order of 10^{-50} m^{-2}) required a specialized optimization routine. The implementation of non-parametric bootstrapping provided rigorous 95% confidence intervals, successfully navigating the discontinuities and mathematical penalty boundaries inherent to the parameter space.
- **Empirical Constraints on Curvature and Thermodynamics:** The analysis directly constrained the expansion parameter (X^2) and the spatial curvature (k). The resulting magnitude for k is on the order of 10^{-52} m^{-2} , which strongly supports the consensus of a nearly absolute spatially flat universe (Weinberg 2008). Crucially, by propagating the empirical uncertainty of these purely kinematic parameters, we independently derived the present-day matter density (ρ_0), vacuum pressure (P), and the Cosmological Constant (Λ). This geometric approach naturally satisfies the dark energy equation of state ($P = -\rho c^2$) and resolves the degeneracy between mass-energy density and the expansion rate.

Future Directions: The analytical solution based on hyperbolic functions provided a highly stable mathematical structure for modeling local extragalactic data. To further test the predictive power of this exact metric, future research should incorporate deeper redshift catalogs—such as those derived from Type Ia supernovae or Baryon Acoustic Oscillations (Weinberg 2008). Extending the empirical fit to higher redshifts will serve to further restrict the confidence intervals of X and k , and evaluate the stability of the derived thermodynamic properties across different cosmic epochs.

In conclusion, this work reaffirms that the geometric symmetry properties of spacetime (Carroll 2019) are not merely static background conditions, but active dynamical constraints capable of dictating the kinematic and thermodynamic evolution of the cosmos.

Data and Code Availability

The raw empirical data analyzed in this study—specifically the extragalactic distances and recession velocities—were extracted from the NASA/IPAC Extragalactic Database (NED) (NASA/IPAC Extragalactic Database (NED), n.d.), operated by the Jet Propulsion Laboratory, California Institute of Technology, under contract with NASA.

To ensure the transparency and reproducibility of our results, all computational routines, data processing scripts, and the cleaned data structures used for the nonlinear least squares (NLS) regression were developed in the R programming language (R Core Team 2023). The complete source code, the R Markdown manuscript file, and the specific data subsets generated during this study are fully accessible in the project's public GitHub repository: <https://github.com/LUCHER4321/flrw-metric>.

Appendices

Units and Definitions

$$\begin{aligned} x^0 &:= ct, \quad x^r := r, \quad x^\theta := \theta, \quad x^\phi := \phi \\ [x^0] &= m, \quad [r] = m, \quad [\theta] = 1, \quad [\phi] = 1 \\ [a] &= 1, \quad [k] = m^{-2} \\ \dot{a} &:= a_{,0}, \quad \ddot{a} := a_{,00}, \quad \ddot{\ddot{a}} := a_{,000} \\ [\dot{a}] &= m^{-1}, \quad [\ddot{a}] = m^{-2}, \quad [\ddot{\ddot{a}}] = m^{-3} \end{aligned}$$

Metric Tensor

$$\begin{aligned} ds^2 &= -c^2 dt^2 + a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right) \\ g_{00} &= -1, \quad g_{rr} = \frac{a^2}{1 - kr^2}, \quad g_{\theta\theta} = a^2 r^2, \quad g_{\phi\phi} = a^2 r^2 \sin^2 \theta, \\ g^{00} &= -1, \quad g^{rr} = \frac{1 - kr^2}{a^2}, \quad g^{\theta\theta} = \frac{1}{a^2 r^2}, \quad g^{\phi\phi} = \frac{\csc^2 \theta}{a^2 r^2} \end{aligned}$$

Christoffel Symbols

$$\begin{aligned} \Gamma_{\beta\gamma}^\alpha &= \frac{1}{2} g^{\alpha\sigma} (g_{\beta\sigma,\gamma} + g_{\gamma\sigma,\beta} - g_{\beta\gamma,\sigma}) \\ \Gamma_{rr}^0 &= \frac{a\dot{a}}{1 - kr^2}, \quad \Gamma_{\theta\theta}^0 = a\dot{a}r^2, \quad \Gamma_{\phi\phi}^0 = a\dot{a}r^2 \sin^2 \theta, \\ \Gamma_{0r}^r &= \Gamma_{r0}^r = \Gamma_{0\theta}^\theta = \Gamma_{\theta0}^\theta = \Gamma_{0\phi}^\phi = \Gamma_{\phi0}^\phi = \frac{\dot{a}}{a}, \\ \Gamma_{rr}^r &= \frac{kr}{1 - kr^2}, \quad \Gamma_{\theta\theta}^r = (kr^2 - 1)r, \quad \Gamma_{\phi\phi}^r = (kr^2 - 1)r \sin^2 \theta, \\ \Gamma_{r\theta}^\theta &= \Gamma_{\theta r}^\theta = \Gamma_{r\phi}^\theta = \Gamma_{\phi r}^\theta = \frac{1}{r}, \\ \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta, \quad \Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = \cot \theta \end{aligned}$$

Riemann Tensor

$$R_{\beta\gamma\mu}^{\alpha} = \Gamma_{\beta\mu,\gamma}^{\alpha} - \Gamma_{\beta\gamma,\mu}^{\alpha} + \Gamma_{\beta\mu}^{\sigma}\Gamma_{\sigma\gamma}^{\alpha} - \Gamma_{\beta\gamma}^{\sigma}\Gamma_{\sigma\mu}^{\alpha}$$

$$\begin{aligned} R_{r0r}^0 &= \frac{a\ddot{a}}{1-kr^2}, \quad R_{\theta0\theta}^0 = a\ddot{a}r^2, \quad R_{\phi0\phi}^0 = a\ddot{a}r^2 \sin^2 \theta, \\ R_{rr0}^0 &= \frac{a\ddot{a}}{kr^2-1}, \quad R_{\theta\theta0}^0 = -a\ddot{a}r^2, \quad R_{\phi\phi0}^0 = -a\ddot{a}r^2 \sin^2 \theta, \\ R_{00r}^r &= R_{00\theta}^{\theta} = R_{00\phi}^{\phi} = \frac{\ddot{a}}{a}, \quad R_{0r0}^r = R_{0\theta0}^{\theta} = R_{0\phi0}^{\phi} = -\frac{\ddot{a}}{a}, \\ R_{\theta r\theta}^r &= R_{\theta\phi\theta}^{\phi} = r^2 (k + \dot{a}^2), \quad R_{\phi r\phi}^r = R_{\phi\theta\phi}^{\theta} = r^2 (k + \dot{a}^2) \sin^2 \theta, \\ R_{\theta\theta r}^r &= R_{\theta\theta\phi}^{\phi} = -r^2 (k + \dot{a}^2), \quad R_{\phi\phi r}^r = R_{\phi\phi\theta}^{\theta} = -r^2 (k + \dot{a}^2) \sin^2 \theta, \\ R_{rr\theta}^{\theta} &= R_{rr\phi}^{\phi} = \frac{k + \dot{a}^2}{kr^2-1}, \quad R_{r\theta r}^{\theta} = R_{r\phi r}^{\phi} = \frac{k + \dot{a}^2}{1-kr^2} \end{aligned}$$

Cosmological Tensor

$$K_{\beta\gamma\mu\nu}^{\alpha} := R_{\beta\gamma\mu;\nu}^{\alpha} = R_{\beta\gamma\mu,\nu}^{\alpha} + \Gamma_{\rho\nu}^{\alpha} R_{\beta\gamma\mu}^{\rho} - \Gamma_{\beta\nu}^{\rho} R_{\rho\gamma\mu}^{\alpha} - \Gamma_{\gamma\nu}^{\rho} R_{\beta\rho\mu}^{\alpha} - \Gamma_{\mu\nu}^{\rho} R_{\beta\gamma\rho}^{\alpha}$$

$$\begin{aligned} K_{r0r0}^0 &= K_{rr00}^0 = \frac{a\ddot{a} - \dot{a}\ddot{a}}{1-kr^2}, \quad K_{\theta0\theta0}^0 = (a\ddot{a} - \dot{a}\ddot{a})r^2, \quad K_{\phi0\phi0}^0 = (a\ddot{a} - \dot{a}\ddot{a})r^2 \sin^2 \theta, \\ K_{\theta r\theta r}^0 &= K_{r\theta r\theta}^0 = \frac{a\dot{a}r^2}{1-kr^2} (k + \dot{a}^2 - a\ddot{a}), \quad K_{\phi r\phi r}^0 = K_{r\phi r\phi}^0 = \frac{a\dot{a}r^2}{1-kr^2} \sin^2 \theta (k + \dot{a}^2 - a\ddot{a}), \\ K_{rr\theta\theta}^0 &= K_{\theta r r r}^0 = \frac{a\dot{a}r^2}{1-kr^2} (a\ddot{a} - k - \dot{a}^2), \quad K_{rr\phi\phi}^0 = K_{\phi r r r}^0 = \frac{a\dot{a}r^2}{1-kr^2} \sin^2 \theta (a\ddot{a} - k - \dot{a}^2), \\ K_{\theta\theta00}^0 &= (\dot{a}\ddot{a} - a\ddot{a})r^2, \quad K_{\phi\phi\theta\theta}^0 = K_{\theta\phi\theta\phi}^0 = a\dot{a}r^4 \sin^2 \theta (k + \dot{a}^2 - a\ddot{a}), \\ K_{\theta\theta\phi\phi}^0 &= K_{\phi\phi\theta\theta}^0 = a\dot{a}r^4 \sin^2 \theta (a\ddot{a} - k - \dot{a}^2), \quad K_{\phi\phi00}^0 = (\dot{a}\ddot{a} - a\ddot{a})r^2 \sin^2 \theta, \\ K_{00r0}^r &= K_{00\theta0}^{\theta} = K_{00\phi0}^{\phi} = \frac{a\ddot{a} - \dot{a}\ddot{a}}{a^2}, \quad K_{\theta0\theta r}^r = K_{r\theta r0}^r = K_{0r\theta\theta}^r = K_{\theta\theta\theta\phi}^{\phi} = K_{\theta\phi0\theta}^{\phi} = K_{0\phi\theta\theta}^{\phi} = \frac{\dot{a}r^2}{a} (a\ddot{a} - k - \dot{a}^2), \\ K_{\phi0\phi r}^r &= K_{r\phi0\phi}^r = K_{0r\phi\phi}^r = K_{\phi0\phi\theta}^{\theta} = K_{\phi\theta0\phi}^{\theta} = K_{0\theta\phi\phi}^{\theta} = K_{\phi\phi0\theta}^{\theta} = K_{0\phi\theta\phi}^{\theta} = \frac{\dot{a}r^2}{a} \sin^2 \theta (a\ddot{a} - k - \dot{a}^2), \\ K_{\theta0r\theta}^r &= K_{\theta\theta0r}^r = K_{0\theta r\theta}^r = K_{\phi\phi0r}^r = K_{\theta0\phi\theta}^{\phi} = K_{0\theta\theta\phi}^{\phi} = K_{\theta\theta0\phi}^{\phi} = \frac{\dot{a}r^2}{a} (k + \dot{a}^2 - a\ddot{a}), \\ K_{\phi0r\phi}^r &= K_{0\phi r\phi}^r = K_{\phi0\theta\phi}^{\theta} = \frac{\dot{a}r^2}{a} \sin^2 \theta (k + \dot{a}^2 - a\ddot{a}), \quad K_{0r00}^r = K_{0\theta00}^{\theta} = K_{0\phi00}^{\phi} = \frac{\dot{a}\ddot{a} - a\ddot{a}}{a^2}, \\ K_{\theta r\theta0}^r &= K_{\theta\phi\theta0}^{\phi} = \frac{2\dot{a}r^2}{a} (a\ddot{a} - k - \dot{a}^2), \quad K_{\phi r\phi0}^r = K_{\phi\theta\phi0}^{\theta} = \frac{2\dot{a}r^2}{a} \sin^2 \theta (a\ddot{a} - k - \dot{a}^2), \\ K_{\theta\theta00}^r &= \frac{2\dot{a}r^2}{a} \sin^2 \theta (k + \dot{a}^2 - a\ddot{a}), \quad K_{\phi\phi00}^r = \frac{2\dot{a}r^2}{a} \sin^2 \theta \sin^2 \theta (k + \dot{a}^2 - a\ddot{a}), \\ K_{r0\theta r}^{\theta} &= K_{0r\theta r}^{\theta} = K_{rr0\theta}^{\theta} = K_{r0\phi r}^{\phi} = K_{0r\phi r}^{\phi} = K_{rr0\phi}^{\phi} = \frac{\dot{a}}{a(1-kr^2)} (k + \dot{a}^2 - a\ddot{a}), \\ K_{r0r\theta}^{\theta} &= K_{r\theta0r}^{\theta} = K_{\theta0rr}^{\theta} = K_{r0r\phi}^{\phi} = K_{r\phi0r}^{\phi} = K_{0\phi rr}^{\phi} = \frac{\dot{a}}{a(1-kr^2)} (a\ddot{a} - k - \dot{a}^2), \\ K_{rr\theta0}^{\theta} &= K_{rr\phi0}^{\phi} = K_{\theta\theta\phi0}^{\phi} = K_{\phi\phi\theta0}^{\theta} = \frac{2\dot{a}}{a(1-kr^2)} (k + \dot{a}^2 - a\ddot{a}), \quad K_{r\theta r0}^{\theta} = \frac{2\dot{a}}{a(1-kr^2)} (a\ddot{a} - k - \dot{a}^2) \end{aligned}$$

Ricci Tensor

$$R_{\alpha\beta} = R_{\alpha\gamma\beta}^{\gamma}$$

$$\begin{aligned} R_{00} &= -\frac{3\ddot{a}}{a}, \\ R_{rr} &= \frac{a\ddot{a} + 2k + 2\dot{a}^2}{1 - kr^2}, \\ R_{\theta\theta} &= r^2 (a\ddot{a} + 2k + 2\dot{a}^2), \\ R_{\phi\phi} &= r^2 \sin^2 \theta (a\ddot{a} + 2k + 2\dot{a}^2) \end{aligned}$$

Ricci Scalar

$$R = R_{\alpha}^{\alpha} = g^{\alpha\beta} R_{\alpha\beta}$$

$$R = \frac{6}{a^2} (a\ddot{a} + k + \dot{a}^2)$$

Cosmological Principle

$$\text{Cosmological Principle} \Rightarrow K_{\beta\gamma\mu\nu}^{\alpha} = 0$$

$$\begin{aligned} k + \dot{a}^2 - a\ddot{a} &= 0 \wedge \dot{a}\ddot{a} - a\ddot{\dot{a}} = 0 \Rightarrow \\ a &= \frac{1}{2X} \left(Y e^{Xct} + \frac{k}{Y} e^{-Xct} \right) \\ \dot{a} &= \frac{1}{2} \left(Y e^{Xct} - \frac{k}{Y} e^{-Xct} \right) = \sqrt{X^2 a^2 - k} \\ \ddot{a} &= \frac{X}{2} \left(Y e^{Xct} + \frac{k}{Y} e^{-Xct} \right) = X^2 a \\ \ddot{\dot{a}} &= \frac{X^2}{2} \left(Y e^{Xct} - \frac{k}{Y} e^{-Xct} \right) = X^2 \sqrt{X^2 a^2 - k} \\ [X] &= m^{-1}, [Y] = m^{-1} \end{aligned}$$

Einstein Tensor

$$G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R$$

$$\begin{aligned} G_{00} &= \frac{3}{a^2} (k + \dot{a}^2) = 3X^2, \\ G_{rr} &= \frac{2a\ddot{a} + k + \dot{a}^2}{kr^2 - 1} = \frac{3X^2 a^2}{kr^2 - 1}, \\ G_{\theta\theta} &= -r^2 (2a\ddot{a} + k + \dot{a}^2) = -3X^2 a^2 r^2, \\ G_{\phi\phi} &= -r^2 \sin^2 \theta (2a\ddot{a} + k + \dot{a}^2) = -3X^2 a^2 r^2 \sin^2 \theta \\ G_{\alpha\beta} &= -3X^2 g_{\alpha\beta} \end{aligned}$$

Pressure & Cosmological Constant

$$\begin{aligned}
G_{\alpha\beta} + \Lambda g_{\alpha\beta} &= \frac{8\pi G}{c^4} T_{\alpha\beta} \\
T_{\alpha\beta} &= \left(\rho + \frac{P}{c^2} \right) u_\alpha u_\beta + P g_{\alpha\beta} \\
u^\alpha e_\alpha &= c e_0 \Rightarrow u_\alpha e^\alpha = -c e^0
\end{aligned}$$

$$\begin{aligned}
(-3X^2 + \Lambda) g_{\alpha\beta} &= \frac{8\pi G}{c^4} \left(\left(\rho + \frac{P}{c^2} \right) u_\alpha u_\beta + P g_{\alpha\beta} \right) \\
\rho &= \frac{c^2}{8\pi G} (3X^2 - \Lambda) \\
P &= \frac{c^4}{8\pi G} (-3X^2 + \Lambda) = -\rho c^2 \\
\Lambda &= 3X^2 - \frac{8\pi G}{c^2} \rho
\end{aligned}$$

Measurements and Calculations

$$\begin{aligned}
D &:= \text{Measured Distance}, [D] = m \\
V &:= \text{Measured Speed}, [V] = \frac{m}{s} \\
ct = 0 &\rightarrow a = a_0 \\
a &= a_0 \cosh Xct + \frac{\dot{a}_0}{X} \sinh Xct \\
D &= \int \sqrt{g_{\alpha r} dx^\alpha dr} = \frac{a}{\sqrt{k}} \sin^{-1}(\sqrt{k}r) \\
V &= \frac{dD}{dt} = cD \sqrt{X^2 - \frac{k}{a^2}} \\
N &:= \frac{2X^2}{k} a_0^2 - 1, [N] = 1 \\
M &:= \frac{2X^2}{k} a_0 \sqrt{a_0^2 - \frac{k}{X^2}}, [M] = 1 \\
V &= cDX \frac{\sinh(2XD) - M}{\cosh(2XD) + N}
\end{aligned}$$

Results

$$\begin{aligned}
X &\in]1.917023 \times 10^{-26}, 2.110307 \times 10^{-26}[m^{-1} \\
k &\in]2.915996 \times 10^{-52}, 3.662952 \times 10^{-52}[m^{-2} \\
\rho &\in]1.933047 \times 10^{-28}, 2.47118 \times 10^{-26}[\frac{kg}{m^3} \\
P &\in]-2.220986 \times 10^{-09}, -1.737336 \times 10^{-11}[\frac{kg}{m \times s^2} \\
\Lambda &\in]6.412729 \times 10^{-52}, 1.332411 \times 10^{-51}[m^{-2}
\end{aligned}$$

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